

1、使用前准备工作

1、安装SDK

解压文件夹里边的YDLidar-SDK-master.zip文件，得到一个文件YDLidar-SDK-master，把这个文件copy到自己的文件目录下，这里以安装在/home目录下为例子，然后依次输入，

```
cd YDLidar-SDK-master
mkdir build
cd build
cmake ..
make
sudo make install
```

2、编译ROS功能包

ROS1

解压文件夹里边的ydlidar_ros_ws.rar文件，得到一个文件ydlidar_ros_ws，把这个功能包copy到自己环境的根目录下，然后在工作空间打开终端，输入以下指令编译，

```
cd ydlidar_ros_ws
catkin_make -DCATKIN_WHITELIST_PACKAGES=ydlidar_ros_driver
source install/setup.bash          #激活工作空间
```

ROS2

解压文件夹里边的ydlidar_ros2_ws.rar文件，得到一个文件ydlidar_ros2_ws，把这个功能包copy到自己环境的根目录下，然后在工作空间打开终端，输入以下指令编译，

```
cd ydlidar_ros2_ws
colcon build --symlink-install
source install/setup.bash          #激活工作空间
```

3、创建串行端口别名

为了防止多个USB设备接入时候识别错误，我们创建模块串行端口别名，

```
sudo chmod 0777 ~/catkin_ws/src/ydlidar_ros_driver/startup/*
sudo sh ~/catkin_ws/src/ydlidar_ros_driver/startup/initenv.sh
```

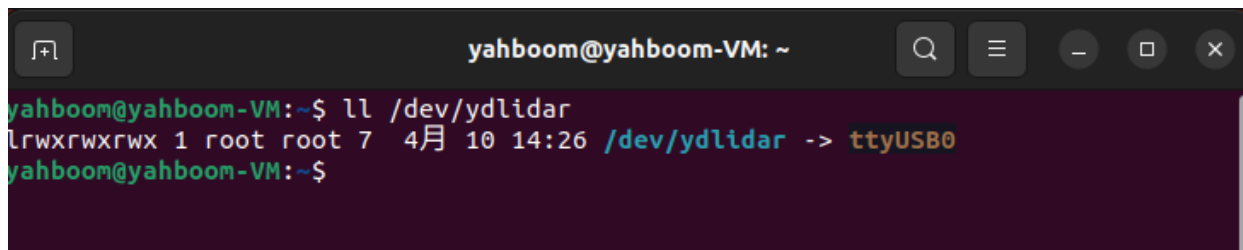
这里的 ~/catkin_ws/src/ydlidar_ros_driver 是安装功能包的位置，需要根据实际情况进行修改。注意：

完成之前的操作后，请再次重新插入模块。

输入以下命令检查是否创建识别成功，

```
ll /dev/ydlidar
```

出现以下画面则表示成功，



```
yahboom@yahboom-VM: ~  
yahboom@yahboom-VM:~$ ll /dev/ydlidar  
lrwxrwxrwx 1 root root 7 4月 10 14:26 /dev/ydlidar -> ttyUSB0  
yahboom@yahboom-VM:~$
```

注意：这里不一定是ttyUSB0，根据系统识别分配到的序号为准。

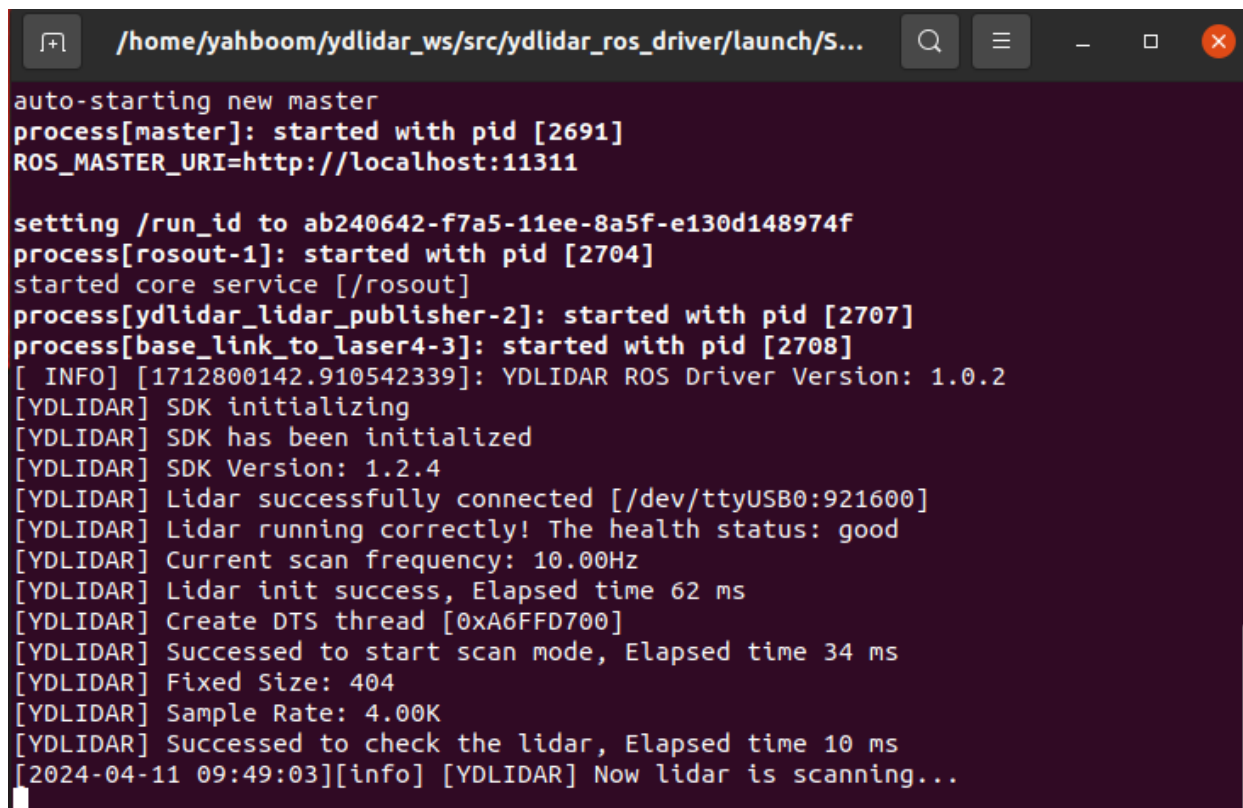
4、运行程序

终端输入，

ROS1,

```
source install/setup.bash  
roslaunch ydlidar_ros_driver SDM18.launch
```

成功运行截图如下，



```
/home/yahboom/ydlidar_ws/src/ydlidar_ros_driver/launch/S...  
auto-starting new master  
process[master]: started with pid [2691]  
ROS_MASTER_URI=http://localhost:11311  
  
setting /run_id to ab240642-f7a5-11ee-8a5f-e130d148974f  
process[rosout-1]: started with pid [2704]  
started core service [/rosout]  
process[ydlidar_lidar_publisher-2]: started with pid [2707]  
process[base_link_to_laser4-3]: started with pid [2708]  
[ INFO] [1712800142.910542339]: YDLIDAR ROS Driver Version: 1.0.2  
[YDLIDAR] SDK initializing  
[YDLIDAR] SDK has been initialized  
[YDLIDAR] SDK Version: 1.2.4  
[YDLIDAR] Lidar successfully connected [/dev/ttyUSB0:921600]  
[YDLIDAR] Lidar running correctly! The health status: good  
[YDLIDAR] Current scan frequency: 10.00Hz  
[YDLIDAR] Lidar init success, Elapsed time 62 ms  
[YDLIDAR] Create DTS thread [0xA6FFD700]  
[YDLIDAR] Succeeded to start scan mode, Elapsed time 34 ms  
[YDLIDAR] Fixed Size: 404  
[YDLIDAR] Sample Rate: 4.00K  
[YDLIDAR] Succeeded to check the lidar, Elapsed time 10 ms  
[2024-04-11 09:49:03][info] [YDLIDAR] Now lidar is scanning...
```

可以输入以下命令来查看具体的数据，

```
rostopic echo /scan
```

```
yahboom@yahboom-virtual-machine: ~/ydlidar_ws
yahboom@yahboom-virtual-machine:~/ydlidar_ws$ rostopic echo /scan
header:
  seq: 1593
  stamp:
    secs: 1712800196
    nsecs: 186082000
  frame_id: "laser_frame"
angle_min: -3.1415927410125732
angle_max: 3.1415927410125732
angle_increment: inf
time_increment: inf
scan_time: 0.029839999973773956
range_min: 0.009999999776482582
range_max: 50.0
ranges: [2.990999937057495]
intensities: [1525.0]
---
header:
  seq: 1594
  stamp:
    secs: 1712800196
    nsecs: 216199000
  frame_id: "laser_frame"
angle_min: -3.1415927410125732
```

其中的，ranges中的数据就是实际测得的距离，单位是米。

ROS2

```
source install/setup.bash
ros2 launch ydlidar_ros2_driver ydlidar_launch.py
```

成功运行出现以下截图，

```
yahboom@yahboom-VM: ~  
yahboom@yahboom-VM:~$ ros2 launch ydlidar_ros2_driver ydlidar_launch.py  
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2024-04-10-14-31-32-285878-yahboom-VM-5767  
[INFO] [launch]: Default logging verbosity is set to INFO  
[INFO] [ydlidar_ros2_driver_node-1]: process started with pid [5778]  
[INFO] [static_transform_publisher-2]: process started with pid [5780]  
[static_transform_publisher-2] [WARN] [1712730692.611677654] []: Old-style arguments are deprecated; see --help for new-style arguments  
[static_transform_publisher-2] [INFO] [1712730692.665684054] [static_tf_pub_laser]: Spinning until stopped publishing transform  
[static_transform_publisher-2] translation: ('0.000000', '0.000000', '0.020000')  
[static_transform_publisher-2] rotation: ('0.000000', '0.000000', '0.000000', '1.000000')  
[static_transform_publisher-2] from 'base_link' to 'laser_frame'  
[ydlidar_ros2_driver_node-1] [INFO] [1712730692.667410531] [ydlidar_ros2_driver_node]: [YDLIDAR INFO] Current ROS Driver Version: 1.0.1  
[ydlidar_ros2_driver_node-1]  
[ydlidar_ros2_driver_node-1] [YDLIDAR] SDK initializing  
[ydlidar_ros2_driver_node-1] [YDLIDAR] SDK has been initialized  
[ydlidar_ros2_driver_node-1] [YDLIDAR] SDK Version: 1.2.4  
[ydlidar_ros2_driver_node-1] [YDLIDAR] Lidar successfully connected [/dev/ttyUSB0:921600]  
[ydlidar_ros2_driver_node-1] [YDLIDAR] Lidar running correctly! The health status: good  
[ydlidar_ros2_driver_node-1] [YDLIDAR] Current scan frequency: 10.00Hz  
[ydlidar_ros2_driver_node-1] [YDLIDAR] Lidar init success, Elapsed time 176 ms  
[ydlidar_ros2_driver_node-1] [YDLIDAR] Create DTS thread [0x44E11640]  
[ydlidar_ros2_driver_node-1] [YDLIDAR] Succeeded to start scan mode, Elapsed time 31 ms  
[ydlidar_ros2_driver_node-1] [YDLIDAR] Fixed Size: 404  
[ydlidar_ros2_driver_node-1] [YDLIDAR] Sample Rate: 4.00K  
[ydlidar_ros2_driver_node-1] [YDLIDAR] Succeeded to check the lidar, Elapsed time 10 ms  
[ydlidar_ros2_driver_node-1] [2024-04-10 14:31:32][info] [YDLIDAR] Now lidar is scanning...
```

可以输入以下命令来查看具体的数据，

```
ros2 topic echo /scan
```

```
yahboom@yahboom-VM: ~  
angle_min: -3.1415927410125732  
angle_max: 3.1415927410125732  
angle_increment: .inf  
time_increment: .inf  
scan_time: 0.04973199963569641  
range_min: 0.009999999776482582  
range_max: 50.0  
ranges:  
- 3.188999891281128  
intensities:  
- 1151.0  
---  
header:  
  stamp:  
    sec: 1712730887  
    nanosec: 197642000  
  frame_id: laser_frame  
angle_min: -3.1415927410125732  
angle_max: 3.1415927410125732  
angle_increment: .inf  
time_increment: .inf  
scan_time: 0.05055199936032295  
range_min: 0.009999999776482582  
range_max: 50.0  
ranges:  
- 3.1600000858306885  
intensities:  
- 1169.0
```

其中的，ranges中的数据就是实际测得的距离，单位是米。

